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United States Patent	12404636
Kind Code	B2
Date of Patent	September 02, 2025
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Spreading machines with improved controls

Abstract

A vehicle for distributing a fluid or a solid material on or into the ground has one or more axles for supporting the vehicle on the ground, a vessel for holding the material to be spread on or into the ground, and a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground. The vehicle may have a plurality of control switches for controlling the lateral extent of distribution of material and operative to control the opening and closing of different valves or different groups of valves of the plurality of valves. The vehicle may have an indicator for each valve or group of valves, the indicator operative to provide different indications for a plurality of different statuses for the corresponding valve or valves.

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Appl. No.:	17/589377
Filed:	January 31, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230243111 A1	Aug. 03, 2023

Related U.S. Application Data

continuation parent-doc US 29824503 20220125 PENDING child-doc US 17589377

Publication Classification

Int. Cl.: E01C19/21 (20060101); E01C19/08 (20060101)

U.S. Cl.:

CPC E01C19/21 (20130101); E01C19/08 (20130101);

Field of Classification Search

CPC: E01C (19/08); E01C (19/21)

USPC: 404/75; 404/104-108; 404/111; 404/118

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 29/824,503 filed Jan. 25, 2022, which is incorporated by reference herein in its entirety.

FIELD OF THE INVENTION

(1) The invention relates to spreading machines, such as asphalt distributors and chip spreaders, having controls for controlling the width and location of spreading.

BACKGROUND

(2) Spreading machines are used in a variety of fields, such as road maintenance and agriculture. The width of spread often has to be adjusted during spread. For example, if a machine is being used to plant seeds in a field that is not rectangular, it is advantageous to be able to adjust the spread to maximize the amount of field planted while minimizing the seeds wasted. Similarly, road maintenance machines such as asphalt distributors and chip spreaders, may be used on roads whose width varies such as in corners, intersections, or with the addition or subtraction of lanes in the road.

(3) To be able to achieve the desired width or placement of material being spread, such machines may have one or more banks of toggle switches. Each toggle switch controls the spreading of a certain segment of the width of spread. An operator of such a machine often has at best an obstructed view of the actual spreading and often relies on the position of the toggle switch to know where the spread is occurring.

(4) However, the installation of these switches is labor-intensive. Additionally, these switches if they break can result in costly downtime. A need therefore exists for an improved bank of switches.

(5) So-called taper switches have begun to become popular with operators. The taper switch allows the width of spread to be adjusted without having to locate the specific toggle switch that controls the specific segment. However, the position of the toggle switch will then not correspond to whether or not the spreading is occurring at the corresponding segment. A need therefore exists for indicating what segments are spreading.

SUMMARY OF THE INVENTION

(6) One embodiment of the invention is a vehicle for distributing a fluid or a solid material on or into the ground. The vehicle can be in the form of a tractor-trailer or a truck, for example. The vehicle has one or more axles for supporting the vehicle on the ground and a vessel for holding a material to be spread on or into the ground. The material can be a fluid or a flowable solid. The vehicle has a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground and a control panel for controlling the lateral extent of distribution of material. The control panel has a plurality of dedicated switches. Each of the dedicated switches has an open position and a closed position. The dedicated switches are operative to respectively control the opening and closing of different valves of the plurality of valves. The dedicated switches are plugged into an electronic bus on a circuit board.

(7) The vehicle may have a plurality of spray nozzles. Each of the valves are in fluid communication with one or more of the spray nozzles. The spray nozzles are arranged laterally to

distribute material across the maximum lateral extent of distribution when all of the valves are open. The vehicle may also have a heater for the vessel and a pump operable to pump liquid from the vessel through the open control valves and out the corresponding nozzles. The heater may be electric or may burn a fuel like propane, gasoline or diesel.

(8) Alternatively, the vehicle may have an elongated hopper having a lateral length to effect the maximum lateral extent of distribution when all of the valves are open. Each valve includes a gate at the bottom of the elongated hopper. The vehicle may also have a conveyor for moving solid material from the vessel to the spread hopper. The vessel may have an open top for receiving solid material from a dump truck or a live bottom trailer. The vessel may also be elongated.

(9) The dedicated switches are replaceable by unplugging an existing dedicated switch and plugging in a new dedicated switch in its place.

(10) The vehicle may have a taper switch operative to increase or decrease the lateral extent of distribution while overriding the dedicated switches. The taper switch could be located in the control panel or elsewhere.

(11) The vehicle may also have an indicator for each of the dedicated switches. The indicators are operative to provide different indications for a plurality of different statuses for the valve or valves corresponding to each dedicated switch. The indicators may be located in the control panel and adjacent the corresponding dedicated switch. An indicator is typically illuminated, and may be a light like an LED, but could instead be a portion of a long LCD strip, typically along a bank of the dedicated switches. Alternatively, it could be particular spots on a screen, typically along a bank of the dedicated switches. The indicators could also be built in the dedicated switches.

(12) The indications may include an indication for material being distributed through the corresponding valve or valves. This indication could be flashing green. Another possible indication is that of the corresponding valve or valves being closed because the dedicated switch is in the off position. This indication could be red. Another indication is that of the corresponding valve or valves being closed, but the corresponding dedicated switch being in the open position. This indication could be amber in color. The vehicle may also have a pump or a conveyor for moving material from the vessel for distribution. The control panel may also have a pump or conveyor switch that has an on position and an off position and is operative to turn the pump or conveyor on or off. Another indication is one when the corresponding dedicated switch is in the on position, but the pump or conveyor switch is in the off position. This indication can be green in color.

(13) The vehicle may also have a taper switch operative to increase or decrease the lateral extent of distribution overriding the dedicated switch positions. Another indication is one indicating the inner most dedicated switch overridden by the taper switch. This indication can be blue in color.

(14) Another embodiment of the invention is a vehicle for distributing a fluid or a solid material on or into the ground. The vehicle can be in the form of a tractor-trailer, a truck, or a specialized vehicle, for example. The vehicle has one or more axles for supporting the vehicle on the ground. The vehicle has a vessel for holding the material to be spread on or into the ground, a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground, and a plurality of control switches for controlling the lateral extent of distribution of material and operative to control the opening and closing of different valves or different groups of valves of the plurality of valves. The control switches may be part of a control panel. The control switches are intended to be used by an operator of the vehicle to control the opening and closing of different valves or different groups of valves of the plurality of valves. The vehicle also has an indicator for each valve or group of valves. The indicators are operative to provide different indications for a plurality of different statuses for the corresponding valve or valves. The number of valves in a group is generally less than 6 and the number of indicators is generally above 6. The number of different indications may be 5 or less, but could exceed 5.

(15) The control switches may include a taper switch operative to increase or decrease the lateral extent of distribution. The increase or decrease changes the status of one or more valves, e.g.,

opens or closes one or more of the valves. Typically, there are two taper switches, one for each side of the vehicle.

(16) The indications for the indicators may include an indication for material being distributed through the corresponding valve or valves, which can be flashing green, and an indication that the corresponding valve or valves are open, but there is no flow of material from the vessel to the valves, such as when a pump or conveyor for moving the fluid or solids from the tank has not been turned on. This latter indication can be solid green. The control switches may include a taper switch operative to increase or decrease the lateral extent of distribution and wherein the indications further comprise an indication indicating the corresponding valve or valves are turned off by operation of the taper switch. This indication may be amber. The indications may include an indication indicating the corresponding valve or valves is the inner most valve or valves turned off by operation of the taper switch. This indication may be blue. The control switches may include a plurality of dedicated switches, each of the dedicated switches having an open position and a closed position which are operative to respectively control the opening and closing of different valves of the plurality of valves. The indications may also include an indication of the corresponding valve or valves being closed because the corresponding dedicated switch is in the off position. This indication may be red.

(17) The vehicle may include spray nozzles. Each of the valves is in fluid communication with one or more of the spray nozzles. The spray nozzles are arranged laterally to distribute material across the maximum lateral extent of distribution when all of the valves are open. Typically the spray nozzles are arranged on a spray bar. The vehicle may have a heater for the vessel and a pump operable to pump liquid from the vessel through the valves that are open and out the corresponding nozzles.

(18) The vehicle may have an elongated hopper having a lateral length to effect the maximum lateral extent of distribution when all of the valves are open. Each valve has a gate at the bottom of the spread hopper. The hopper may be extendable. The vehicle may have a conveyor for moving solid material from the vessel to the hopper. The vessel may have an open top for receiving solid material from a dump truck or a live bottom trailer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a fluid distributor. Wings should be 2 feet long with 6 sprays and center bar should be 8 ft wide with 24 valves.

(2) FIG. 2 is a front view of a control panel of the fluid distributor of FIG. 1.

(3) FIG. 3 is a schematic exploded side view of a module of the control panel of FIG. 2 taken along line 3-3.

(4) FIG. 4 is a side view of a solids distributor.

(5) FIG. 5 is a front view of the solids distributor of FIG. 4.

(6) FIG. 6 is a magnified view of a control valve shown in FIG. 5.

(7) FIG. 7 is a front view of a control panel of the solids distributor of FIG. 4.

DETAILED DESCRIPTION OF THE INVENTION

(8) In a first embodiment of the invention, a fluid distributor **10** is provided. Fluid distributor **10** includes a motorized vehicle **12** (shown in phantom), typically a truck, having a cab **14**, one or more side view mirrors **15**, two or more axles **16**, a chassis **18**, and a tank or vessel **20** mounted to chassis **18**. Tank **20** is used to store the fluid to be distributed. Tank **20** may be heated by a heater **21** of any suitable kind. Heater **21** is shown schematically as an electric heater. The fluid may be gas, liquid or combinations thereof. The liquid may be asphalt, water, aqueous solutions, slurries, coatings, or emulsions. In a preferred embodiment, the liquid is asphalt and fluid distributor **10** is

an asphalt distributor, an example of which is described in U.S. Pat. No. 4,817,870, which is incorporated herein by reference.

(9) Fluid distributor **10** has an engine **22** (shown schematically), which drives a transmission **24** (shown schematically) and a power-take-off drive **28** off of transmission **24**, which in turn drives a hydraulic drive pump **30**, which in turn drives hydraulic motor **32**, which in turn drives fluid pump **34** via gear box **36**. Fluid pump **34** need not be driven as described. Any suitable drive can be used such as an electric motor for pump **34**. Pump **34** may be of any suitable type. Preferably, it is of the constant displacement type. Fluid pump **34** draws fluid from tank **20** via an intake port **38** having an intake valve **40**. For simplicity, FIG. **1** does not show additional ports such as a return port that are conventional.

(10) Fluid pump **34** pumps fluid from tank **20** to a fluid distribution system **42** located at the rear end of vehicle **12**. Working from the end backwards, fluid distribution system **42** has spray nozzles **44** (or outlets) for spraying the fluid, control valves **46**, spray bar **48**, and conduit/piping system means **50**. Spray nozzles **44** are arranged on spray bar **48**, typically uniformly spaced along spray bar **48**. Instead of spray nozzles **44**, any suitable applicator for applying fluid may be used. For example, the applicator can be an injector such as those used to inject ammonia into soil for agricultural applications or to spray nutrient solutions. Spray bar **48** may be extendable or fixed. Fixed spray bars have a length that typically is approximately the same as or narrower than vehicle **12**. Extendable spray bars of any suitable design are designed to extend beyond the width of vehicle **12** in operation, but to retract to be at or less than the width of vehicle **12** for ease of transportation. As shown, spray bar **48** is extendable having pivotal end portions **48a** and **48b** and a central portion **48c**. Pivotal end portions are sometimes called spray bar wings. Alternatively, spray bar **48** could be a variable width spray bar having central portion **48c** and end portions that move in and out in several positions. End portion **48a** is shown in the extended position with spraying occurring in FIG. **1** while end portion **48b** is shown in the retracted position with no spraying occurring. Central portion **48c** is shown spraying. Spray bar **48** may be located at the front end or the rear of vehicle **12**. Flow of fluid to spray nozzles **44** is controlled by control valves **46**. In one embodiment, there are three spray nozzles **44** per foot of spray bar **48** and one control valve **46** per spray nozzle **44**. Other arrangements are also contemplated, such as one valve **46** controlling the flow through three spray nozzles **44**. Fluid distribution system **42** has the necessary conduit, whether tubing, hoses, piping or mixtures thereof for connecting distribution system **42** together.

(11) Fluid distributor **10** has a control panel **100** for controlling the distribution of fluid. Control panel **100** is typically located in cab **14** so that the operator of fluid distributor **10** can drive vehicle **12** while controlling the distribution of fluid via control valves **46**.

(12) Control panel **100** as shown in FIG. **2** has a master power switch **102**, switches **104** and **106** for controlling spray bar wings **48a** and **48b**, indicators **105** and **107**, switches **108** and **110** for controlling the extension and retraction of end portions of a variable width spray bar **48**, a joystick **112**, and a screen **114**. Master power switch **102** turns on and off power to control panel **100**. Switches **104** and **106** raise and lower spray bar wings **48a** and **48b**, respectively. Spray bar wings **48a** and **48b** raise so that they can be out of the way when not in use and lower so they can be used for spraying fluid into the ground. There are multiple switches **104** and **106** in case end portions **48a** and **48b** having more than one segment. Indicators **105** and **107** correspond to switches **104** and **106** and indicate whether end portions **48a** and **48b** are raised or lowered. Joystick **112** allows spray bar **48** to be moved up, down, right and left as needed. Screen or display **114** can be used to show various operating parameters and available functions. In addition, the screen allows for the operator to input application specific settings for items such as flow and speed. Also, the screen allows for the display of machine diagnostics. Display **114** can display a rear view from a rear view camera (not shown).

(13) Control panel **100** has a master spray switch **120**, dedicated spray switches **122** and **124** arranged in banks **125a** and **125b**, indicators **126** and **128** (also arranged in banks **125a** and **125b**),

taper switches **130** and **132**, and empty slots **134**. Master spray switch **120** operates to turn on and off fluid pump **34**. Dedicated spray switches **122** in bank **125a** and spray switches **124** in bank **125b** have on and off positions that operate to turn on and off control valves **46** on the left and right side of vehicle **12**, respectively. Switches **122a**, **122b**, **124a** and **124b** are shown in the on position. The other switches **122** and **124** are shown in the off position. One switch **122** or **124** can operate one or more valves **46**. In a preferred embodiment, one switch **122** or **124** controls three valves **46** and each valve **46** is connected to one spray nozzle **44** (note that FIGS. **1** and **2** do not necessarily have a matched number of switches to valves).

(14) Indicators **126** and **128** are located immediately above dedicated spray switches **122** and **124** and provide indications representing the status of control valves **46**. For example, indicator **126a** provides an indication of the status of the one or more control valves **46** controlled by switch **122a** and indicator **128a** provides an indication of the status of the one or more control valves **46** controlled by switch **124a**. Indicator **126** and **128** can be any suitable kind of indicator that lets an operator of fluid distributor **10** know the status of control valves **46**. In a preferred embodiment, indicators **126** or **128** are multicolor LEDs, one for each switch **122** and **124**, to show at least some of the different statuses using different colors. Alternatively, an elongated LED or LCD display (not shown) could have multiple portions that constitute indicators **126** and **128**. As shown each indicator **126** and **128** is located directly above and in registry with corresponding switches **122** and **124**. It is contemplated that indicators **126** and **128** can be located in a different location (outside of control panel **100**) or that there can be duplicate indicators **126** and **128** located not adjacent to switches **122** and **124**. For example, indicators **126** or duplicate indicators **126** can be located in side view mirror **15a** on the left side and indicators **128** or duplicates can be located in side view mirror **15** on the right side such that an operator can look in one of side view mirrors **15** to check the spraying and also see the indications. The indicators **126** and **128** that are located (or duplicated) in side view mirrors **15** may be limited to those that correspond to the spray nozzles whose sprays can be seen in the side view mirrors **15**. Similarly, indicators **126** and **128** or duplicates may be located in display **114** when display **114** shows views from a rear view camera in which case switches **122** and **124** may also be located in registry along display **114**.

(15) Taper switches **130** and **132** are operative to override the dedicated switches **122** and **124** to turn on and off control valves **46**. In one alternative, taper switches **130** and **132** can only override only the dedicated switches that are in the on position. Conceivably, the overriding could be partial. In other words, if each dedicated switch **122** or **124** controls three control valves **46**, taper switches **130** and **132** could override dedicated switches **122** or **124** as to one control valve **46** at a time thereby achieving individualized control over control valves **46**. In another alternative, taper switches **130** and **132** can override the dedicated switches regardless of the position they are in. Taper switches **130** and **132** are preferably rocker switches, which return to a central position after being pushed.

(16) FIG. **3** is an exploded cross-sectional view of bank **125b**. Bank **125a** and bank **125b** are substantially identical and are preferably just different halves of the same bank **125**. Bank **125b** includes switches **124**, a faceplate **136** shared with bank **125a**, a printed circuit board **138**, and a case **140**. Switch **122/124** is preferably a maintained rocker or toggle switch having a lever **142** and electrical male contacts **144**. Switches **122** and **124** are plugged/press fit into printed circuit board **138** having female contacts **146** for male contacts **144**. Board **138** has a controller area network (CAN) bus. Preferably bank **125** is available as a CAN module. Advantageously, bank **125** has a number of slots for accommodating a typical maximum number of switches **122/124** for ease of manufacturing. Preferably, bank **125** has 8 or more slots (note that CAN modules having more than 4 slots were not previously commercially available), more preferably 18-30 slots, and most preferably 24 slots. In practice, this means that bank **125** has switches **122** and **124** and may have one or more empty slots **134**. Bank **125** advantageously eliminates the need to individually wire each switch **122/124** as in the prior art. Furthermore, if a switch **122** or **124** breaks, it is a simple

matter for an operator to replace the broken switch by unplugging it and plugging in a new one. Previously, the operator would have had to unsolder the broken switch and solder in the new switch often at great delay.

(17) In operation, operator of fluid distributor **10** first prepares distributor **10** for distributing the fluid in vessel **20**. Depending on the width of the road onto which asphalt is to be sprayed on, end portions **48a** or **48b** may be lowered (fully or partially) by switches **104** and **106** for spray wings or be extended by switches **108** and **110** for extendable spray bars. Next, switches **122** and **124** can be flipped to the on position or off position to match the desired width of spraying, but spraying has not started because master spray switch **120** is in the off position. Indicators **126** and **128** corresponding to switches **122** and **124** that are in the on position are solid green or are amber if the switches are overridden. Switches **122** and **124** may be overridden because the corresponding spray nozzles **44** are not in position for spraying, i.e., the spray nozzles are on an end portion **48a** or **48b** that has not been lowered or extended. Switches **122** and **124** may also be overridden by operation of taper switches **130** and **132**, respectively, discussed later. The remaining indicators are off if there is no corresponding switch **122** or **124**.

(18) Once preparations are complete, the operator can drive distributor **10** and then turn on master spray switch **120** at the appropriate time to start spraying. At this time, all indicators **126** and **128** that were solid green are now flashing green indicating that the fluid is being sprayed out corresponding spray nozzles **44**. Indicators that were off or amber continue to be off or amber. No fluid is being distributed through nozzles **44** that correspond to indicators **126** and **128** that are amber. The operator can adjust the spray width as necessary by feel, specifically, the operator can feel switches **122** and **124** and from their position, on or off, and location determine which ones need to be changed.

(19) Alternatively, the operator can use one or both taper switches **130** and **132** to adjust the spray width. Hypothetically, assume that switches **122a-122f** are in the on position and indicators **126a-126f** are flashing green indicating that fluid is being spray through the corresponding spray nozzles **44**, while switches **122g** and **122h** are in the off position and indicators **126g** and **126h** are red indicating that there is no spraying through the corresponding spray nozzles **44**. If the operator wants to reduce the extent of spray on the left side of fluid distributor **10**, the operator can press taper switch **130** to the right once. This results in indicator **126f** turning blue and the corresponding control valves **46** closing. The blue of indicator **126f** indicates that spray switch **122f** is the inner most switch **122** overridden by operation of taper switch **130**. If the operator wants to reduce the extent of spray on the left side of fluid distributor **10** again, the operator can press taper switch **130** to the right. This results in indicator **126e** turning blue and indicator **126f** turning amber and the corresponding control valves **46** being closed. Taper switch **130** can continue to be pressed to the right until indicators **126b-126f** are all amber and indicator **126a** is blue in which case control valves **46** for the left side of distributor **10** will be closed. Pushing taper switch **130** to the right does not affect the status of valves **46** controlled by control switches **122g** and **122h**; also indicators **126g** and **126h** remain red. Taper switch **130** can be pressed left to reverse the effect of pressing taper switch to the right. Taper switch **130** can be pressed left causing the indicator that is blue “to move left” like a cursor until the indicators **126** for all the switches **122** that are in the on position are green. Taper switch **132** works the same with respect to control switches **124** and indicators **128** for the right side of distributor **10**. Indicators **126** and **128** therefore provide easy to understand statuses for control valves **46** and spray nozzles **44**.

(20) Conceivably, if taper switches **130** allow individualized control over valves **46** assuming switches **122** do not allow individualized control, indicators **126** could show a hybrid status of green (or flashing green) and blue when the corresponding switch **122** has been overridden as to some control valves **46**, but not others.

(21) EXAMPLE OF FLUID DISTRIBUTOR—In a preferred example, fluid distributor **10** is used to spray asphalt onto a road. Distributor **10** has a heater **21** for keeping asphalt liquid. Spray bar **48**

when fully extended is approximately 24 feet wide, each portion **48a**, **48b** and **48c** being eight foot long. Each portion **48a**, **48b** and **48c** has 24 spray nozzles **44** connected to 24 control valves **46**. Bank **125** has 12 switches **122** and 12 switches **124**. In other words each switch **122** or **124** has three corresponding spray nozzles **44** and three corresponding control valves **46**.

(22) In a second embodiment of the invention, a solids distributor **210** is provided. Solids distributor **210** includes a motorized vehicle **212** having optionally a cab **214**, two or more axles **216**, a chassis **218**, and a vessel **220** mounted to chassis **218**. Vessel **220** is used to hold the solids to be distributed. Vessel **220** may be open top (as shown) or closed top. Vessel **220** may be heated by a heater of any suitable kind. The solids may vary in size from a fine powder, pellets, to coarse aggregates and may include a variety of different sizes. The solids may be seeds, fertilizer, aggregate, gravel, sand or combinations thereof, for example. In a preferred embodiment, the solids are gravel or aggregate and solids distributor **10** is a chip spreader, an example of which is described in U.S. Pat. No. 5,234,128, which is incorporated herein by reference.

(23) Solids distributor **210** has a conveyor system **222** for moving the solids from vessel **220** to an elongated hopper **224** (or vessel). Conveyor system **222** (shown schematically) may be made of any suitable conveyor or combinations thereof including screw conveyors, belt conveyors, bucket conveyors, vibrating conveyors, pneumatic conveyors, and combinations thereof. Conveyor system can include parallel conveyors as shown schematically in FIG. 5 for better distributing solids into spread hopper. Hopper **224** receives the solids conveyed by conveyor system **222**. Hopper **224** has optionally a vibrator **226** (shown schematically) to better distribute solids within hopper **224** and a plurality of control valves **228** located at the bottom of hopper **224**. Control valves **228** may be any suitable valve for the solids to be distributed. As shown, control valves **228** are of the gate variety having a gate **230** and a cylinder **232** having a piston **234**. Extension and retraction of piston **234** causes gate **230** to pivotally close and open, respectively, over an outlet **244** of hopper **224**.

Alternatively, gate **230** can be a sliding gate. Cylinder **232** may be pneumatic or hydraulic. Hopper **224** has outlets **244** at the bottom, one outlet **244** for each valve **228**. Outlets **244** may be continuous such that they are not distinct from each other, but are separate only to the extent that they are closed by different valves **228**. Outlets **244** as shown are upstream of valves **228**, but can be downstream.

(24) Elongated hopper **224** has a width L in its most compact form for transportation purposes. It may be extended on both sides usually by same amount to have widths L1, L2 or L3 to match the desired lateral extent of distribution. It may be necessary depending on the design of hopper **224** to remove or install control valves **228** to change the width or lateral length of hopper **224**.

(25) Solids distributor **210** has a control panel **300**. Control panel has an emergency stop button **302**, a display power button **304** and a display **306**. The purpose of display power button **304** is to turn on and off display **306**. Display **306** is preferably a touch screen display permitting an operator to control application settings such as type of solid to be spread, solid size (e.g., gravel size), ground speed, and application rate. Display **306** can also display operating parameters, available functions, and machine diagnostics. Control panel **300** also has dedicated spreading switches **322** and **324** arranged in banks **325a** and **325b**, respectively, and corresponding indicators **326** and **328**, respectively. Dedicated spreading switches **322** and **324**, banks **325a** and **325b**, and indicators **326** and **328**, may be the same or similar to dedicated spray switches **122** and **124**, banks **125a** and **125b**, and indicators **126** and **128**. Control panel **300** also has taper switches **332** that are the same or similar to taper switches **132**. Bank **325** may have empty slots (not shown) like empty slots **134** for bank **125**.

(26) In operation, operator of fluid distributor **210** first prepares distributor **210** for distributing the solids in vessel **220**. This may include adjusting the length of vessel to meet or exceed the maximum desired width of spreading, say L1, for example. Next, switches **322** and **324** can be flipped to the on position or off position to match the desired width of spreading. Indicators **326** and **328** corresponding to switches **322** and **324** that are in the on position are solid green or are

amber if the switches are overridden. Switches **322** and **324** may be overridden because vessel **220** may not have been elongated sufficiently that the corresponding gates **230** are not in position for spraying, i.e., control valves **228** shown on either end of vessel **224** for lengths **L2** and **L3** are not in position to be used. Switches **322** and **324** may also be overridden by operation of taper switches **330** and **332**, respectively, discussed later. The remaining indicators **326** and **328** are off if there is no corresponding switch **322** or **324**.

(27) Once preparations are complete, the operator can drive solids distributor **210** and then turn on the appropriate controls via touch screen **306** at the appropriate time to start spraying. At this time, all indicators **326** and **328** that were solid green are now flashing green indicating that the solids are being distributed out of corresponding valves **228** and outlets **244**. Indicators **326** and **328** that were off or amber continue to be off or amber. No solids are distributed through outlets **244** that correspond to indicators **326** and **328** that are amber. The operator can adjust the spray width as necessary by feel, specifically, the operator can feel switches **322** and **324** and from their position, on or off, and location determine which ones need to be changed.

(28) Alternatively, the operator can use one or both taper switches **330** and **332** to adjust the distribution width. Hypothetically, assume that switches **322a-322f** are in the on position and indicators **326a-326f** are flashing green indicating that solids are being distributed spray through outlets **244**, while switches **322g** and **322h** are in the off position and indicators **326g** and **326h** are red indicating that there is no distribution through the corresponding outlets **244**. If the operator wants to reduce the extent of spray on the left side of solids distributor **210**, the operator can press taper switch **330** to the right once. This results in indicator **326f** turning blue and the corresponding valves **228** closing. The blue of indicator **326f** indicates that spray switch **322f** is the inner most switch **322** overridden by operation of taper switch **330**. If the operator wants to reduce the extent of spray on the left side of solids distributor **210** again, the operator can press taper switch **330** to the right. This results in indicator **326e** turning blue and indicator **326f** turning amber and the corresponding control valves **228** being closed. Taper switch **330** can continue to be pressed to the right until indicators **326b-326f** are all amber and indicator **326a** is blue in which case control valves **228** for the left side of distributor **210** will be closed. Pushing taper switch **330** to the right does not affect the status of valves **228** controlled by control switches **322g** and **322h**; also indicators **326g** and **326h** remain red. Taper switch **330** can be pressed left to reverse the effect of pressing taper switch to the right. Taper switch **330** can be pressed left causing the indicator that is blue “to move left” like a cursor until indicators **326** for all the switches **322** that are in the on position are green. Taper switch **332** works the same with respect to control switches **324** and indicators **328** for the right side of distributor **210**. Indicators **326** and **328** therefore provide easy to understand statuses for control valves **228** and outlets **244**.

(29) **EXAMPLE OF SOLIDS DISTRIBUTOR**—In a preferred example, solids distributor **210** is used to spread chips onto a road. Vessel **220** has an open top for receiving aggregate from a dump truck **T** or a live bottoms trailer. Elongated hopper **224** is approximately 24 feet wide when fully extended and has 24 outlets **244** and 24 control valves **228**. Bank **325** has 12 switches **322** and 12 switches **324**. In other words, each switch **322** or **324** has one corresponding outlet **244** and one corresponding control valve **228**.

(30) While the invention has been described with respect to certain embodiments, as will be appreciated by those skilled in the art, it is to be understood that the invention is capable of numerous changes, modifications and rearrangements, and such changes, modifications and rearrangements are intended to be covered by the following claims.

Claims

1. A vehicle for distributing a fluid or a solid material on or into the ground, the vehicle comprising: one or more axles for supporting the vehicle on the ground; a vessel for holding the

material to be spread on or into the ground; a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground; and a control panel for controlling the lateral extent of distribution of material, the control panel having a plurality of dedicated switches, each of the dedicated switches having an open position and a closed position, which are operative to respectively control the opening and closing of different valves of the plurality of valves, the dedicated switches plugged into an electronic bus on a circuit board, the vehicle further comprising a taper switch operative to control the opening of an outermost closed valve of the plurality of valves and the closing of an outermost open valve of the plurality of valves to incrementally increase or decrease the lateral extent of distribution while overriding the dedicated switches, wherein repeated operation of the taper switch to increase the lateral extent repeatedly increasing the lateral extent of distribution until a maximum lateral extent of distribution is achieved on one side of the vehicle.

2. The vehicle of claim 1 further comprising a plurality of spray nozzles, each of the valves in fluid communication with one or more of the spray nozzles, the spray nozzles being arranged laterally to distribute material across the maximum lateral extent of distribution when all of the valves are open.

3. The vehicle of claim 2 further comprising a heater for the vessel and a pump operable to pump liquid from the vessel through the open control valves and out the corresponding nozzles.

4. The vehicle of claim 1 further comprising an indicator for each of the dedicated switches, the indicator operative to provide different indications for a plurality of different statuses for the valve or valves corresponding to each dedicated switch.

5. The vehicle of claim 4 wherein the indications comprise an indication for material being distributed through the corresponding valve or valves, an indication of the corresponding valve or valves being closed by operation of the dedicated switch is in the off position, and an indication of the corresponding valve or valves being closed, but the corresponding dedicated switch is in the open position.

6. The vehicle of claim 4 further comprising a pump or a conveyor for moving material from the vessel for distribution, wherein the control panel further comprises a pump or conveyor switch having an on position and an off position and operative to turn the pump or conveyor on or off and the indications further comprise an indication that the corresponding dedicated switch is in the on position, but the pump or conveyor switch is in the off position.

7. The vehicle of claim 4 wherein the indications further comprise an indication indicating the inner most dedicated switch overridden by the taper switch.

8. A vehicle for distributing a fluid or a solid material on or into the ground, the vehicle comprising: one or more axles for supporting the vehicle on the ground; a vessel for holding the material to be spread on or into the ground; a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground; and a control panel for controlling the lateral extent of distribution of material, the control panel having a plurality of dedicated switches, each of the dedicated switches having an open position and a closed position, which are operative to respectively control the opening and closing of different valves of the plurality of valves, the dedicated switches plugged into an electronic bus on a circuit board, the vehicle of further comprising an elongated hopper having a lateral length to affect the maximum lateral extent of distribution when all of the valves are open, each valve comprising a gate at the bottom of the elongated hopper.

9. The vehicle of claim 8 further comprising a conveyor for moving solid material from the vessel to the spread hopper.

10. The vehicle of claim 9 wherein the vessel has an open top for receiving solid material from a dump truck or a live bottom trailer.

11. A vehicle for distributing a fluid or a solid material on or into the ground, the vehicle comprising: one or more axles for supporting the vehicle on the ground; a vessel for holding the

material to be spread on or into the ground; a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground; and a control panel for controlling the lateral extent of distribution of material, the control panel having a plurality of dedicated switches, each of the dedicated switches having an open position and a closed position, which are operative to respectively control the opening and closing of different valves of the plurality of valves, the dedicated switches plugged into an electronic bus on a circuit board, wherein the dedicated switches are replaceable by unplugging an existing dedicated switch and plugging in a new dedicated switch in its place.

12. A vehicle for distributing a fluid or a solid material on or into the ground, the vehicle comprising: one or more axles for supporting the vehicle on the ground; a vessel for holding the material to be spread on or into the ground; a plurality of valves for controlling the lateral extent of distribution of the material on or into the ground; a plurality of control switches for controlling the lateral extent of distribution of material and operative to control the opening and closing of different valves or different groups of valves of the plurality of valves; and an indicator for each valve or group of valves, wherein there are a plurality of indicators, the indicators operative to provide different indications for a plurality of different statuses for the corresponding valve or valves, the indications comprising an indication for material being distributed through the corresponding valve or valves, and an indication that the corresponding valve or valves are open, but there is no flow of material from the vessel to the valves.

13. The vehicle of claim 12 wherein the control switches further comprise a taper switch operative to increase or decrease the lateral extent of distribution and wherein the indications further comprise an indication indicating the corresponding valve or valves are turned off by operation of the taper switch.

14. The vehicle of claim 13 wherein the indications further comprise an indication indicating the corresponding valve or valves is the innermost valve or valves turned off by operation of the taper switch.

15. The vehicle of claim 13 wherein the control switches comprise dedicated switches, each of the dedicated switches having an open position and a closed position which are operative to respectively control the opening and closing of different valves of the plurality of valves, wherein the indications further comprise an indication of the corresponding valve or valves being closed because the corresponding dedicated switch is in the off position.

16. The vehicle of claim 12 further comprising a plurality of spray nozzles, each of the valves in fluid communication with one or more of the spray nozzles, the spray nozzles being arranged laterally to distribute material across the maximum lateral extent of distribution when all of the valves are open.

17. The vehicle of claim 16 further comprising a heater for the vessel and a pump operable to pump liquid from the vessel through the valves that are open and out the corresponding nozzles.

18. The vehicle of claim 12 further comprising an elongated hopper having a lateral length to affect the maximum lateral extent of distribution when all of the valves are open, each valve comprising a gate at the bottom of the spread hopper.

19. The vehicle of claim 18 further comprising a conveyor for moving solid material from the vessel to the hopper.

20. The vehicle of claim 19 wherein the vessel has an open top for receiving solid material from a dump truck or a live bottom trailer.
